3.6.2 Nervous coordination (A-level only)

3.6.2.1 Nerve impulses (A-level only)

Content	Opportunities for skills development
The structure of a myelinated motor neurone. The establishment of a resting potential in terms of differential membrane permeability, electrochemical gradients and the movement of sodium ions and potassium ions. Changes in membrane permeability lead to depolarisation and the generation of an action potential. The all-or-nothing principle. The passage of an action potential along non-myelinated and myelinated axons, resulting in nerve impulses. The nature and importance of the refractory period in producing discrete impulses and in limiting the frequency of impulse transmission. Factors affecting the speed of conductance: myelination and saltatory conduction; axon diameter; temperature.	**MS 0.2** Students could use appropriate units when calculating the maximum frequency of impulse conduction given the refractory period of a neurone.

3.6.2.2 Synaptic transmission (A-level only)

Content	Opportunities for skills development
The detailed structure of a synapse and of a neuromuscular junction. The sequence of events involved in transmission across a cholinergic synapse in sufficient detail to explain: • unidirectionality • temporal and spatial summation • inhibition by inhibitory synapses. A comparison of transmission across a cholinergic synapse and across a neuromuscular junction. **Students should be able to** use information provided to predict and explain the effects of specific drugs on a synapse. (Recall of the names and mode of action of individual drugs will **not** be required.)	